

Charles "Chaz" Merritt

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Education

University of Georgia, BA in Cognitive Science	2020 – 2024
- GPA: 3.5/4.0 Coursework: Search & Optimization, Learning Algorithms, Epistemology	
University of Georgia, MS in Artificial Intelligence	2024 – 2026
Coursework: Reinforcement Learning, Data Science, Advanced Forest Planning Research: Quantifying Economic Value-at-Risk in Timber Management via Markov Chain Monte Carlo (MCMC)	

Experience

Research Assistant, University of Georgia – Athens, GA	Aug. 2024 – May 2026
- Contributed to peer-reviewed papers at the intersection of AI and forestry. - Built a full-stack web interface for optimization workflows (Python, Django, TypeScript) - Applied Monte Carlo simulation and risk modeling (Value-at-Risk, CVaR) to quantify economic uncertainty	
Geospatial Analyst Intern, Northwest Management – Moscow, ID	May – Sep. 2024
- Designed automated GIS pipelines using Python, Rasterio, and GeoPandas. - Implemented clustering algorithms and image segmentation models (SAM) for forest analysis. - Processed and analyzed large-scale spatial datasets for operational forestry decision-making.	
OMNI Intern, Oak Ridge National Lab – Oak Ridge, TN	May – Dec. 2023
- Supported development of intrusion prevention systems for enterprise networks. - Gained hands-on experience with SmartNICs (Nvidia BlueField DPUs). - Presented to stakeholders at a Department of Energy poster session in Washington, D.C.	

Projects

Optimization Software Web Interface	2026
- Developed an interactive webapp for the SETS timber supply models. - Tools Used: Railway, Django, Python, TypeScript, HiGHS-Py	
Brandon AI	2026
- Developed contrastive representation models for phonetic similarity learning with InfoNCE loss. - Tools Used: PyTorch, NumPy	

Publications

Thesis: Quantifying Economic Value-at-Risk in Timber Management via Markov Chain Monte Carlo (MCMC)	May. 2026
Chaz Merritt, Chair: Pete Bettinger	
Peer-Reviewed Article: AI Technology Use in Forestry, Journal of Forestry	Feb. 2026
Krista Merry, Pete Bettinger, Chaz Merritt, Khaled Rasheed, Bruno da Silva, Roger C. Lowe III	

Skills

Programming: Python, TypeScript
Machine Learning: PyTorch, Reinforcement Learning, Contrastive Learning, MCMC, Optimization
Data & GIS: QGIS, ArcGIS, GeoPandas, Rasterio, NumPy, Pandas